CVR COLLEGE OF ENGINEERING

(An UGC Autonomous Institution with NAAC 'A' Grade, Affiliated to JNTUH)
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DEPARTMENT OF CSE(DATA SCIENCE)

COURSE OUTCOMES (COs)
OF
B. TECH. **CSE (DATA SCIENCE)**

R25 REGULATIONS

I Year I Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
25BS101	**Matrices and Calculus**	CO1	Find rank of a matrix and solve a linear system of equations
		CO2	Evaluate eigen values, eigen vectors and diagonalize the given matrix
		CO3	Apply Mean value theorems to solve engineering problems and to trace the cartesian curves
		CO4	Find the extremum of a multi-variate function with/without constraint
		CO5	Evaluate multiple integrals and apply the concept to find the areas and volumes
25BS102/152	**Advanced Engineering Physics**	CO1	Analyze crystal structures, identify defects, and apply XRD and SEM technique for material characterization
		CO2	Explain the principles of lasers and fibre optics and their applications in communication and sensing
		CO3	Apply quantum mechanical principles to explain particle behaviour and energy band formation in solids
		CO4	Understand quantum computing concepts, use quantum gates, and explain basic quantum algorithms
		CO5	Classify magnetic and dielectric materials and explain their properties, synthesis, and applications

25ME102/152	**Computer Aided Engineering Graphics**	CO1	Apply standard conventions to construct various engineering curves using AutoCAD.
		CO2	Develop orthographic projections of points and lines in any position using the fundamentals of projection theory in AutoCAD.
		CO3	Generate orthographic projections of regular solids in any given position using AutoCAD.
		CO4	Create sectional views and surface developments of basic 3D objects using AutoCAD.
		CO5	Produce isometric views and convert them into multi-view orthographic drawings of simple 3D objects using AutoCAD.
25CY101	**Essentials of System and Web Interfacing**	CO1	Describe computer components, OS functions, and UNIX basics.
		CO2	Work with UNIX file system, shell commands, and permissions.
		CO3	Use filters, pipes, and regular expressions for text processing.
		CO4	Create web pages with HTML elements, forms, and multimedia.
		CO5	Design responsive web pages using CSS layouts and animations.
25CS101	**Programming for Problem Solving**	CO1	Design solutions to simple problems using basic programming concepts realizing appropriate data types and control statements.
		CO2	Solve real world problems using modular constructs and pointers.
		CO3	Implement solutions to complex problems using recursion and homogeneous data types.
		CO4	Analyze, differentiate and implement elementary algorithms of sorting, searching and work on problems using strings.
		CO5	Represent real world data using user defined data types and files to solve problems of relevance.
25HS103/153	**Indian Knowledge System**	CO1	Understand nature, scope and related fields of Indian knowledge system.
		CO2	Recognize the scientific literature available in ancient Indian traditions

		CO3	Understand the application of Bharatiya Jnana Parampara
		CO4	Understand Indian approach towards Wellbeing
		CO5	Appreciate vast contribution of ancient Indian researchers, engineers, scientists and architects to the modern world
25BS131/181	**Advanced Engineering Physics Lab**	CO1	Able to identify the type of crystal system based on X-ray diffractogram.
		CO2	Determine the wavelength of light by diffraction principle and learn methods to minimize the signal loss in optical fibers.
		CO3	Understand the applications of the Photoelectric effect, develop skills to identify the type of semiconductors and determine charge carrier concentration in it using Hall effect.
		CO4	Able to understand the operation of quantum gates using a simulator.
		CO5	Gain knowledge of applications of dielectric materials and hysteresis behavior of magnetic materials.
25CS131	**Programming For Problem Solving Lab**	CO1	Execute commands in Linux programming environment and translate given algorithms to a working and correct program.
		CO2	Interpret syntax errors as reported by the compilers and correct logical errors encountered at run time using debuggers like GDB.
		CO3	Write iterative as well as recursive programs using modular constructs and pointers wherever applicable.
		CO4	Apply Algorithms for solving problems like sorting, searching and on strings.
		CO5	Represent data in Structures and Unions and manipulate data in files through a program.
25CY131	**Essentials of Systems and Web Interfacing Lab**	CO1	Assemble computer components and configure dual operating systems.
		CO2	Apply UNIX commands for file management, filters, links, and permissions.
		CO3	Prepare professional documents, perform mail merge and create presentations.
		CO4	Use Excel for statistical analysis, data cleaning, and visualization.
		CO5	Build interactive and responsive websites using HTML and CSS.

25ME131/181	**Engineering Workshop**	CO1	Perform basic engineering trade operations such as carpentry, fitting, and tin smithy with accuracy and proper workmanship.
		CO2	Interpret and adhere to workshop safety regulations to ensure safe working practices.
		CO3	Identify, select, and use marking-out tools, hand tools, and measuring equipment to achieve prescribed tolerances.
		CO4	Select suitable equipment and consumables for specific applications and apply trade knowledge in practical problem-solving situations.
		CO5	Work effectively both individually and collaboratively to plan, execute, and complete workshop tasks.

I Year II Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
25BS151	**Ordinary Differential Equations and Vector Calculus**	CO1	Solve the first-order O.D.E and extend the knowledge to the applications in engineering problems.
		CO2	Solve higher-order O.D.E and extend the knowledge to the applications in engineering problems.
		CO3	Make use of Laplace transform techniques to solve O.D.Es.
		CO4	Evaluate directional derivative of scalar functions and divergence, curl of vector functions.
		CO5	Evaluate the line, surface and volume integrals and translating them from one to another.
25BS154/104	**Engineering Chemistry**	CO1	to understand the fundamental properties of water and its applications in both domestic and industrial purposes.
		CO2	to gain basic knowledge of electrochemical processes and their relevance to corrosion and its control methods.

		CO3	to comprehend the significance and practical applications of batteries and various energy sources, enhancing their potential as future engineers and entrepreneurs.
		CO4	to apply the principles of polymers in real world applications.
		CO5	to apply the knowledge of lubricants, smart materials and biosensors in analyzing industrial and biomedical applications.
25HS151/101	**English for Skill Enhancement**	CO1	Choose appropriate vocabulary and write grammatically correct sentences.
		CO2	Comprehend and compose ideas coherently.
		CO3	Handle official correspondence independently.
		CO4	Condense and paraphrase ideas for various purposes
		CO5	Use language for multiple technical tasks.
25DT151	**Python Programming**	CO1	Apply fundamental Python programming concepts, including syntax, variables, data types, operators, and control structures, to solve basic computational problems.
		CO2	Utilize Python data structures such as lists, tuples, sets, and dictionaries to store, manipulate, and process data efficiently.
		CO3	Implement Python functions, modules, and packages to create reusable, modular, and maintainable code.
		CO4	Perform file handling and exception handling to enable robust and error-tolerant Python programs.
		CO5	Apply data analysis and OOP principles to design scalable applications.
25CS151	**Data Structures**	CO1	Design and implement linear list data structures.
		CO2	Implement stack and queue data structures and their applications.
		CO3	Assimilate the terminology of trees and implement binary tree operations in C.
		CO4	Represent graphs and implement their traversal techniques.

		CO5	Apply Hashing techniques and Files concepts to solve real world problems.
25HS152/102	**Environmental Science**	CO1	Define and understand the structure, function, and significance of ecosystems, including energy flow, biogeochemical cycles
		CO2	Evaluate biodiversity at genetic, species, and ecosystem levels, its values, threats, and conservation methods under national and international frameworks.
		CO3	Analyze the classification, utilization, and sustainable management of natural resources, along with alternative energy options.
		CO4	Identify types, sources, and impacts of environmental pollution, and apply suitable control technologies while assessing protocols.
		CO5	Interpret environmental policies, legislation, and the EIA process to propose management plans addressing Sustainable development goals (SDGs).
25BS182/132	**Engineering Chemistry Lab**	CO1	Determine important parameters such as water hardness and the corrosion rate of mild steel under various conditions.
		CO2	apply the princiles of conductometry to determine concentrations in acid-base reactions.
		CO3	apply potentiometry technique to verify Nernst equation and to determine concentrations.
		CO4	apply techniques like pH metry to determine equivalence points in acid-base reactions.
		CO5	apply the colorimetric technique to verify Beer's law and preparation of a polymer.
25CS181	**Data Structures Lab**	CO1	Implement basic data structures such as arrays, and linked lists.
		CO2	Solve complex problems using data structures such as stacks and queues.
		CO3	Implement operations on Binary Trees and Binary Search Trees
		CO4	Solve problems involving graphs.
		CO5	Use Hash Tables for designing solutions to real world problems

25HS181/131	**English Language and Communication Skills Lab**	CO1	Enunciate / articulate English speech sounds accurately.
		CO2	Comprehend and converse confidently in global scenarios.
		CO3	Distinguish between accents and interpret various visual texts and ideas.
		CO4	Remember and reproduce ideas efficiently.
		CO5	Gauge non-verbal clues and respond appropriately.
25DT181	**Python Programming Lab**	CO1	Develop the application specific codes using python.
		CO2	Understand Strings, Lists, Tuples and Dictionaries in Python.
		CO3	Develop programs using modular approach, file I/O, Python standard library.
		CO4	Implement Digital Systems using Python.
		CO5	Understand and perform data analysis with Python libraries like NumPy and Pandas and grasp the fundamentals of object-oriented programming.

II Year I Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
25BS203	**Mathematical and Statistical Foundations**	CO1	Apply the number theory concepts to cryptography domain
		CO2	Apply the concepts of probability and distributions to some case studies.
		CO3	Correlate the material of one unit to the material in other units
		CO4	Resolve the potential misconceptions and hazards in each topic of study
		CO5	Fit the curve, correlation and regression for the given data
25AM201	**Computer Organization and Architecture**	CO1	Understand the Fundamentals of Digital Computer Organization and Architecture.

		CO2	Understand the operations of Instruction codes, Computer Instructions and Microprogrammed Control.
		CO3	Understand Register and Stack Organization, Addressing modes, Data on Representation as per IEEE Standards.
		CO4	Analyze Computer Arithmetic Operations and Input-Output Organization.
		CO5	Appreciate the Importance of Memory Organization, different types of Memory and Multiprocessors, Pipeline and Vector Processing.
25AM202	**Java Programming**	CO1	Understand basic concepts of Java Programming Language.
		CO2	Identify the model of Object-Oriented Programming: Abstract data types, Encapsulation, Inheritance and Polymorphism.
		CO3	Summarize the fundamental features like Abstract Classes, Interfaces, Packages and Exceptions.
		CO4	Apply and Analyse the advantages of Multi-Threading.
		CO5	Understand the importance of balanced trees and Generic Collections.
25CY202/252	**Software Engineering**	CO1	Explain the fundamental concepts of software engineering and compare various software process models.
		CO2	Analyze and document software requirements by applying requirements engineering activities such as elicitation, feasibility analysis, specification, validation, and requirements management.
		CO3	Apply software design principles and modeling techniques to develop architectural, structural, and behavioral models using UML diagrams.
		CO4	Evaluate software quality through appropriate testing strategies and metrics, including black-box and white-box testing.
		CO5	Assess risks and implement quality management practices by using risk identification, projection, mitigation strategies (RMMM), and applying software quality assurance standards such as ISO 9000.

25DT201	**Database Management Systems**	CO1	Explain fundamental DBMS concepts, architectures, data models, and ER design principles.
		CO2	Apply relational algebra operations and write correct SQL queries for data retrieval and manipulation.
		CO3	Analyse functional dependencies and perform normalization to obtain well-structured database schemas.
		CO4	Evaluate transaction schedules and apply concurrency control techniques to ensure reliable database processing.
		CO5	Evaluate and implement indexing methods such as hash-based and tree-based indexing to optimize query performance in large-scale databases.
25BS231/281	**Computational Mathematics Lab**	CO1	Develop the code to find the Eigen values and Eigen vectors using MATLAB
		CO2	Develop the code to find the solutions of Algebraic and Transcendental equations using MATLAB
		CO3	Develop the code to find the solutions of linear system of equations using iterative methods using MATLAB
		CO4	Write the code to solve problems of first order ODEs using MATLAB
		CO5	Write the code to solve problems of higher order linear differential equations with constant coefficients using MATLAB
25AM231	**Java Programming Lab**	CO1	Apply Object-Oriented programming concepts.
		CO2	Design the applications using Inheritance, Polymorphism.
		CO3	Handle exceptions at Compile time and Run time, apply io stream.
		CO4	Implement Multitasking and Synchronization using Multithreading concepts.
		CO5	Implement Data Structures and Real-World applications using Collection framework.
25CY231/281	**Software Engineering Lab**	CO1	Formulate clear and concise problem statements by analyzing functional and non-functional requirements for software projects.
		CO2	Develop complete Software Requirement Specifications (SRS), design documents,and testing-related documentation following standard software engineering practices.

		CO3	Apply Risk Management principles to prepare structured and traceable project documents.
		CO4	Use design-phase CASE tools to model system architecture and effectively perform software design activities.
		CO5	Design and execute test cases for unit testing, integration testing, and various white-box and black-box testing techniques to ensure software quality.
25DT231	**Database Management Systems Lab**	CO1	Analyze database requirements and determine the entities involved in the system and their relationship to each other.
		CO2	Design E-R Model to represent database application scenarios.
		CO3	Convert/transform the E-R Model to relational tables, populate relational database and formulate SQL queries on data.
		CO4	Improve the database design by normalization.
		CO5	Implement PL/SQL procedures, function, triggers and cursors.
25DT232	**Data Visualization -Tableau**	CO1	Understand How to import data into Tableau.
		CO2	Understand Tableau concepts of Dimensions and Measures.
		CO3	Develop Programs and understand how to map Visual Layouts and Graphical Properties.
		CO4	Create a Dashboard that links multiple visualizations.
		CO5	Use graphical user interfaces to create Frames for providing solutions to real world problems.

II Year II Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
25IT251/201	**Discrete Mathematics**	CO1	Understand and construct precise mathematical proofs
		CO2	Apply logic and set theory to formulate precise statements

		CO3	Analyze and solve counting problems on finite and discrete structures
		CO4	Describe and manipulate sequences
		CO5	Apply graph theory in solving computing problems
25IT253	**Operating Systems**	CO1	Explain the core components of an OS and how it manage resources during user interaction.
		CO2	Understand how CPU scheduling techniques, deadlock handling methods are applied.
		CO3	Identify synchronization issues and apply appropriate inter-process communication (IPC) methods to resolve them.
		CO4	Understanding fundamental memory management techniques and virtual memory concepts.
		CO5	Demonstrate effective file system operations using access methods, directory structures, allocation, protection and essential system calls.
25DT251	**Algorithms Design and Analysis**	CO1	Apply space and time complexity analysis using asymptotic notations & analyse graph traversal algorithms
		CO2	Design divide-and-conquer algorithms and critically assess their runtime and space trade-offs.
		CO3	Define Optimization problems and solve them using greedy strategy.
		CO4	Solving optimization problems using dynamic programming paradigm and backtracking methods.
		CO5	Analyse and design branch-and-bound algorithms, understand NP-hard & NP-Complete problems.
25EE254/205	**Basic Electrical And Electronics Engineering**	CO1	Analyze DC and AC circuits using KVL, KCL, theorems, and basic network methods.
		CO2	Explain construction, operation, and performance of DC machines.
		CO3	Explain working and characteristics of transformers and induction motors.
		CO4	Analyze rectifiers, filters, Zener and varactor diode applications..
		CO5	Explain transistor biasing, CE amplifier, and LT switchgear components.
25DT252	**Introduction to Data Science**	CO1	Explain the data science lifecycle, KDD process, and diverse applications across industries.

		CO2	Preprocess raw data by handling missing values and noise, reduce dimensionality for effective analysis, and identify appropriate data distributions.
		CO3	Create insightful visualizations and dashboards to explore data and communicate analytical findings.
		CO4	Build, evaluate, and compare classification, prediction models.
		CO5	Discover frequent item sets and generate association rules, and understand clustering models for pattern discovery.
25HS201/251	**Innovation And Entrepreneurship**	CO1	Understand the entrepreneurship and the entrepreneurial process and its significance in economic development.
		CO2	Assess the problem from an industry perspective and generate solutions using the design thinking principles.
		CO3	Assess market competition, estimate market size, and develop a prototype.
		CO4	Analyze Business and financial planning models and Go-to-Market strategies.
		CO5	Able to build a start-up, register IP and identify funding opportunities.
25CS281	**Algorithms and Operating Systems Lab**	CO1	Understand and apply Divide and Conquer and Greedy methods for problem solving.
		CO2	Appreciate solutions for real world problems using Dynamic Programming
		CO3	Implement Backtracking method for solving real world problems
		CO4	Simulate operating system concepts such as scheduling, deadlock management, file management and memory management.
		CO5	Implement C programs for handling files and processes using Unix system calls.
25EE284/234	**Electrical and Electronics Engineering Lab**	CO1	Verify and analyze electrical circuit theorems such as superposition, voltage division, and current division in resistive networks.

		CO2	Conduct experiments on DC machines and single-phase transformers to determine their performance parameters, efficiency, and regulation.
		CO3	Measure and evaluate RMS and average values of AC signals, and interpret their practical significance
		CO4	Analyze the operation of rectifier circuits (half-wave, full-wave, and bridge) with and without filters, and study Zener and varactor diode applications for voltage regulation.
		CO5	Design and test transistor biasing circuits and study the frequency response and characteristics of common-emitter amplifiers and LT switchgear components.
25DT281	**Data Science Lab**	CO1	preprocess and transform raw data for effective model development.
		CO2	implement and evaluate data mining techniques for pattern discovery.
		CO3	create meaningful visualizations to communicate analytical insights.
		CO4	build and validate predictive models using suitable ML algorithms.
		CO5	interpret results and make data-driven decisions from model outputs.
25IT282/231	**MongoDB**	CO1	Understand the installation, configuration, and basic operation of MongoDB on local and cloud platforms.
		CO2	Illustrate the creation and management of databases, collections, and documents using suitable MongoDB data types.
		CO3	Apply and analyze CRUD operations, query operators, projections, sorting, and filtering for effective data retrieval.
		CO4	Design and evaluate advanced MongoDB operations such as aggregation pipelines,index ing, and geospatial queries.
		CO5	Implement database interactions using MongoDB Atlas, Compass, and PyMongo for real time application development.